

Ostrom and the eVSM

Where a theory of collective-action institutions and a model of any viable institution overlap, and what each supplies that the other lacks

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The question

Ostrom's model, as Eric Jones redrew it in 2003, is about what lets a collective-action institution come into being, last, and perform. The eVSM (extended Viable System Model) is about what any institution must do to stay viable. If both are right, they should overlap heavily. This note sets out where, using two sources: Jones's figure as redrawn in the RCN Graph Tool (*Ostroms Collective Action in 3 phases*) and Marc's *eVSM plus environment for RCN*.

A later section sets this work beside Jon Walker's own VSM analysis of Ostrom, which reached many of the same conclusions from her eight design principles rather than from Jones's figure.

The short answer: the overlap is large. It sits mostly in Ostrom's **Survival** phase, partly in **Performance**, and there is almost none in **Origin**. The VSM assumes the system already exists; Origin is about whether it forms at all.

The overall layout already matches

Ostrom / Jones	eVSM
Left panel: the users and the resource	The operating units (OU-1 ... OU-n, the Operating Unit Hub) and their local environments
Center panel: the institution's practicalities	The metasystem: spheres S-1 to S-11
Right panel: external factors	The environment column (Regulators, Partners, Competitors, Financiers, Ideal Futures ...)

Jones puts Common Understanding and Decision-making Costs in the *users'* panel, not the institution's. In a commons the users do their own steering. Beer says the same: the metasystem is a set of functions, not a separate class of people. Ostrom's design principle 3 (the people affected by the rules help make them) is what that looks like in practice.

Three positions settled in discussion

1. Trust, Similarities, Experience and Open Communication live on the eVSM's edges

Matching nodes to nodes, these four looked weakly covered. They are not properties of any sphere; they are properties of the channels. Every sphere-to-sphere link in the eVSM runs both ways (33 of the 55 possible sphere pairs are linked, 50 two-way pairs in all), and the quality of those two-way links is where Ostrom's social variables sit:

- **Experience** is Ostrom's "mutually beneficial transactions over time": the history of a two-way channel settling into balance.
- **Trust** reduces how much each side must check the other. A trusted channel does not need auditing.
- **Open Communication** is how much a channel can carry.
- **Similarities** is how little translation a channel needs where two parts meet (Beer's transducers).

RCN's own instruments already read these edges: the Agree / Disagree / Unknown edge ratings and the Block G "who you work with" network in the eVSM measures specification.

The specific channels that carry them between users are the **informal OU-to-OU links**, and that is where most of the variety is handled. The current eVSM drawing links OU-1 to OU-2 and OU-2 to OU-n but not OU-1 to OU-n. Marc will redraw it with an informal channel between every pair of operating units.

Two limits. **Asset Structure** is about how holdings are divided, not about a relationship, so it belongs with Resources & Accounting (S-7), in who pays and who receives. And **Origin** comes before the edges exist: Ostrom's left panel describes how channels get built; the VSM describes channels that are working.

2. Coordination and monitoring: combine the watching, not the sanctioning

Ostrom supplies the argument for folding monitoring into Coordination (S-1). In working commons the monitors are the users themselves or answer to them (design principle 4). That makes monitoring a sideways, peer activity with the same shape as coordination. Jones's own causal chain runs the same way: Info on Others' Resource Use → Common Understanding ↔ Conflict Resolution.

The condition is to combine the *watching*, not the *sanctioning*. Sanctions stay with Managing (S-6) and Policies & Legal (S-10). Beer kept coordination (System 2) and audit (System 3*) apart for a reason: operations share information freely with a coordinator who serves them, and stop once the coordinator is also the inspector. If peers collude and nobody reports, Ostrom's backstop is Dual Enforcement, with Regulators outside.

Beer describes two levels of coordination: operating units coordinating directly with each other, and above them a coordination function that System 3 runs. That second level is an **S-1 to S-6 link**, which the current drawing does not have. It is needed; how to draw it is still open.

Open: how to show the inspection function without losing the simplicity, symmetry and readability of the eVSM metasytem. Jon Walker's analysis (below) separates monitoring into kinds that sit in different places, and suggests most of it can be drawn as edges rather than a new sphere. Nothing is chosen yet.

3. Ostrom's three levels map onto the eVSM's color bands

Ostrom level	eVSM spheres
Constitutional	S-10 Policies & Legal, S-11 Worldview & Identity (magenta)
Bridge	S-8 Planning, S-9 Research (purple): System 4, in both levels
Collective choice	S-5 Innovation, S-6 Managing, S-7 Resources & Accounting (red): System 3, where the budget is made, "taxes" collected and redistributed

Operational

The operating units, S-1 Coordination (teal), and the shared services S-2 HR, S-3 Infrastructure / Tools, S-4 Outreach (blue), which bridge System 3 and the operating units

- **Each band writes the rules for the band below.** In Ostrom, operational rules are made in the collective-choice arena, so S-6 and S-7 make them. The constitutional band decides who sits in the collective-choice arena and how its budget rules can change.
- **Why System 4 sits in both:** constitutional change is usually set off by what Research sees, a changing environment or collective choices that keep failing. That is Jones's Rate of Change → Survival link. Planning/Research also sits in a balancing pair with Managing, and that pair is itself a collective-choice arena.
- **Recursion:** every operating unit has its own metasystem and its own three levels. The larger system's collective-choice rules limit the local system's constitutional choices. That is Ostrom's nesting seen through Beer's recursion.

The Markov-blanket reading

Marc reads the eVSM as a Markov blanket that minimizes free energy (Friston). The sensory / motor states are S-4, S-7 and S-9; the tuning layer is S-1, S-2, S-5, S-10 and S-11; S-3, S-6 and S-8 are the internal core, the layer the eVSM introduction already calls the "internal core layer." Every operating unit is a smaller blanket of its own, one level of recursion down. The drawings mark each node's blanket role.

The blanket holds in the current file: every environment node connects only to S-4, S-7 or S-9. That is also why Regulators should reach Policies & Legal through Research rather than directly. Internal states may not touch the outside.

Ostrom fits this reading closely:

- **Common Understanding**, in Jones's words "information storage with a comparator inside it," is a prior plus a prediction-error signal, the core of free-energy minimization.
- **Ostrom's revised rule**, "the smallest and simplest set of rules relevant to the variety in the environment," is Ashby's Law of Requisite Variety, and it is free energy's trade-off between complexity and accuracy: the simplest model that still fits the world.
- **Rate of Change → Survival** is a world that changes faster than the model can update, so prediction error cannot be brought down.
- **Jones's claim that outside factors reach origin and performance only indirectly** is the blanket condition. His government arrows cross the boundary at a few points: Legal Status, de facto vs. de jure, Dual Enforcement, Technology for Monitoring and Speed of Access to Info. Those are Ostrom's sensory states. Where Jones draws government reaching into the users' panel, the blanket reading says the arrow lands on a sensory state first.

Node by node

strong same function or condition · partial overlapping · **channel** carried by the informal OU-to-OU edges, not by a sphere

Ostrom node	eVSM counterpart		Why
Outside			
Governmental Agencies	Regulators	strong	The most organized outside actor; reaches the eVSM through Research (S-9).
Exogenous Changes	Competitors, Market Geographies	partial	Shocks arrive through the environment column.
Rate of Change	S-9 Research	strong	Beer defines viability as adapting as fast as the environment changes.
Larger System of Institutional Support	The next recursion up; Partners	strong	Nested enterprises (design principle 8) are Beer's recursion.
External Factors	The environment column as a whole	partial	A heading, not a single node.
Constitutional			
Recognized Membership	S-11 Worldview & Identity	strong	Clear boundaries (design principle 1): who is inside, where the blanket is.
Common Understanding	S-11; also S-9, S-8	strong	Store plus comparator: the shared model, prior plus error.
Legal Status, Local Prerogative	S-10 Policies & Legal	strong	Legal standing, handled with Regulators through S-9.
de facto vs. de jure	S-10 Policies & Legal	strong	Whether the rules in use are the rules the state recognizes.
Bridge			
Perceived Alternative Strategies	S-8 Planning, S-5 Innovation	partial	The options the group can see, and the ones it can invent.
Info about Resource	S-9 Research, S-7 Resources & Accounting	partial	Sensing the resource; S-7 as measurement.
Collective choice			
Small, Simple Set of Rules	S-6 Managing; bounded by S-10	strong	Requisite variety. The strongest single match, the hinge between the two models.
Enforcement Costs	S-6 Managing; S-7	strong	Sanctioning is decided here and paid from the budget.
Decision-making Costs	S-6 Managing	strong	The cost of running the collective-choice arena. Group Size raises it: more variety for the metasystem to absorb.

Asset Structure	S-7 Resources & Accounting	strong	Who pays and who receives: the budget and its redistribution.
Dual Enforcement	S-6 Managing, S-10 Policies & Legal	partial	Local enforcement backed from outside; the backstop when peer monitoring fails.
Operational: functions			
Conflict Resolution	S-1 Coordination	strong	Cheap, routine dispute handling is System 2: damping oscillation between units.
Info on Others' Resource Use	S-1 Coordination; the OU-to-OU channels	strong	Peer monitoring: the watching, not the sanctioning.
Technology for Monitoring Resources	S-3 Infrastructure / Tools	strong	Instruments that make the resource observable are shared tools.
Speed of Access to Info	S-3; S-1	partial	The delay, not the data, is the variable.
Operational: the users			
Trust, Open Communication, Similarities, Experience	The informal OU-to-OU channels	channel	Properties of edges, not spheres (position 1 above).
Interdependence / subtractability	OU-to-OU channels; Suppliers (OU)	channel	Units drawing on the same resource have overlapping local environments, which is why they oscillate and need coordination.
Group Size	Operating Unit Hub	partial	How many operating units the metasystem must absorb.
Distance to Resource, Resource Scarcity	Suppliers (OU)	partial	The operating units' shared local environment.
Outcomes			
Survival	Viability	strong	The same concept.
Performance	Viability	partial	What viability buys; Beer measures it with actuality, capability and potentiality.
Origin	None		The VSM assumes the system exists.

What each supplies the other

What Ostrom gives the eVSM

- **Origin:** how the channels get built before they can carry anything.
- **Equity:** Asset Structure puts who gains and who pays into the budget question at S-7.
- **The state as a source of legitimacy,** not just another item in the environment. In the eVSM it arrives through S-9.
- **Dual enforcement** as the backstop behind peer monitoring.

What the eVSM gives Ostrom

- **Balancing the present against the future** (Beer's System 3–System 4 pair). Jones's map has nothing that plans or innovates; Perceived Alternatives is its only forward-looking node.
- **Separating coordination from management.** Ostrom merges them into "rules plus enforcement."
- **Synergy:** System 3's job of making the whole produce more than its parts. Jones's map has no path for gains: Performance is fed only by Decision-making Costs, Enforcement Costs and Survival, so the map explains how a commons avoids losing value, not how it creates it. Ostrom's own word for it is *provision* (see Walker, below).
- **Alarm signals** (Beer's algedonic channel) that go straight to the top. Ostrom's map has no fast path for alarms.
- **Recursion stated outright.** Ostrom implies it with a single node.
- **Blanket discipline:** every outside influence lands on a sensory state first.

Jon Walker's reading of Ostrom

Walker's section "The Commons and the Work of Elinor Ostrom" (7.2.5 of a longer work; it refers to "the authors" and cites Espinosa & Walker 2013, so it is probably written with Angela Espinosa) tests Ostrom's eight design principles against Beer's criteria for viability: all five systems present, connected correctly, and with enough capacity.

What he finds

- **Direct matches:** nested enterprises (principle 8) are recursion; clear boundaries (1) are the system in focus; rules fitted to local conditions (2) are the VSM's environmental niche; recognized self-determination (7) is operational autonomy.
- **System 1** is the users together with the resource and their equipment.
- **System 2** is conflict resolution (principle 6), informal ("the local coffee house") and formal (Valencia's water court). He adds further System 2s: shared values, time-sharing of maintenance, and social control.
- **System 3:** graduated sanctions (5) are its enforcement side. Its main job, creating synergy, is not among the principles, though it runs through Ostrom's cases.
- **System 4** is not among the principles either. His clearest case is the Los Angeles West Basin water producers commissioning an engineer to survey the threat. He proposes adding it as a principle.
- **System 5** is the meeting of all users and its constitution.
- **Only System 2 is explicit** in Ostrom's list.
- **How the systems connect:** autonomy is limited only by the cohesion of the whole. The resource bargain is a closed loop: System 3 allocates resources and the users send accountability information back. Graduated sanctions are Beer's intervention rules. Systems 3, 4 and 5 usually meet as one general meeting of all users, and because the users take part they identify with the whole commons, which he thinks explains high compliance despite low sanctions.
- **His result** is a table of 11 VSM principles for long-lasting commons. Three have no Ostrom counterpart: System 3 works from a real-time model of the operations; System 4 is present; Systems 3 and 4 interact continuously.

Where he confirms this work

Nesting as recursion, boundaries as identity, conflict resolution as System 2, and Ostrom lacking a forward-looking System 4 all match. His general meeting is our point that in a commons the users are the metasytem. His principle that System 3 works from a real-time model built from the users' collective knowledge matches Jones's Common Understanding, the store with a comparator inside it. Two independent readings reach the same map.

Synergy

Beer gives System 3 two jobs. One, keeping each unit within the rules through monitoring and sanctions, is in Ostrom's principles. The other, making the whole produce more than the units could separately, is not. Coordination (System 2) *prevents loss*: it damps oscillation so units don't undo each other's work, as in Alanya's rotation of fishing spots. Synergy (System 3) *creates gain*: something no unit could have alone.

- **Tibi dam:** Alicante's farmers built it together in 1594, and it doubled their water.
- **Torbel:** grazing the alpine pasture in common beat fencing it into plots. Privatizing it would have added fences, markets for grass, and duplicated water systems, so part of the synergy is simply what the commons avoids spending.
- **West Basin:** the producers pooled money for a survey no single producer would have paid for.

Ostrom's vocabulary covers this under another name. Principle 2 speaks of *appropriation and provision* rules. Appropriation rules say who may take what; provision rules say who contributes labour and materials to build and maintain the resource. Provision is her word for synergy, and it links the two frameworks more directly than Walker's text does.

In the eVSM, synergy runs through the collective-choice band and then becomes infrastructure. Managing (S-6) sees the whole set of operating units and finds combinations that pay. Resources & Accounting (S-7) has two halves: redistribution is the resource bargain, and pooling for a shared investment is synergy. Innovation (S-5) proposes joint projects. Once built, a dam is infrastructure: the shared services (S-2, S-3, S-4) are synergy made permanent, which is why they sit between System 3 and the operating units. In Ashby's terms synergy often reduces variety: one dam instead of a hundred wells, one survey instead of twenty guesses.

It also bears on legitimacy. System 3 may limit autonomy only to get a greater payoff for the whole, so the payoff has to be visible. If users cannot see what they gain from the constraints, the constraints read as control rather than coordination. One eVSM question for a neighbourhood group: "*What do we have that none of us could have alone?*" If nobody can name anything, collective choice is running on enforcement alone.

Three kinds of monitoring

Walker places monitoring in two places; Beer adds a third.

	1. Peer watching	2. Accountability	3. Audit (Beer's System 3*)
Direction	Sideways, among equals	Upward: the return leg of the resource bargain	Downward and occasional: System 3 looks directly, bypassing the reports
Who	The users themselves, or monitors who answer to them (principle 4 as Ostrom states it). The Alanya fishermen.	Each operating unit reports what it did with what it was given.	Dedicated monitors who go and look. Valencia's ditch-riders fit here better than under accountability.
eVSM location	The informal OU-to-OU channels and Coordination (S-1). Agreed.	OU → Hub → Managing (S-6) and Resources & Accounting (S-7). Missing today: the Hub → OU edges run one way.	Could be a property of the accountability edge rather than a node.
Purpose	Keeps mutual adjustment honest; feeds Common Understanding and Conflict Resolution.	Closes the loop: resources keep flowing as long as accountability flows back.	Checks that the reports are true.
Powers	None of its own: it carries information, not sanctions.	Triggers the intervention rules (graduated sanctions) held by S-6 and S-10.	Informs System 3; sanctions stay where they are.
Runs on	Trust between equals, and builds it.	The relationship to the metasystem.	The metasystem's need to know independently.

Keep the kinds apart. If peer channels start carrying reports upward, users stop talking freely on them; that is why only the watching was combined into Coordination, not the sanctioning. Drawn this way, peer watching is already agreed, accountability is a set of missing return edges (which keeps the eVSM's pairing and symmetry), and only audit might need something new, possibly just a marking on the accountability edge.

Where this work goes further

- **Ostrom's three levels.** Walker does not use them. He does say that when Systems 3, 4 and 5 are one meeting, everything works through that meeting, and that separate bodies need more design. For a small group run as one meeting, what separates the levels is which rules the meeting is changing, not which body meets.
- **Origin and trust.** He works from the 1990 principles, not Jones's variables, so he says nothing about origin. His nearest point, that participation builds identification with the whole, is a tuning-layer effect in blanket terms and fits the reading of trust as a property of the edges.
- **The Markov-blanket reading** is not in his text.

Two slips in his text: he cites principle 8 where he means principle 7 (self-determination), and the table does not survive extraction from the PDF cleanly, though its three "not stated" rows are clear.

Changes to the eVSM

- **Agreed:** an informal channel between every pair of operating units (Marc will redraw).
- **Agreed:** an S-1 to S-6 link for Beer's second level of coordination. How to draw it is still to decide.
- **Agreed:** Coordination carries the watching; sanctions stay in S-6 and S-10.
- **Not needed:** a direct S-10 to Regulators edge. S-9 carries that traffic, as the blanket requires.
- **Open:** how to draw inspection without breaking the metasystem's symmetry. Leading candidate, after Walker: accountability as return edges $OU \rightarrow Hub \rightarrow S-6 / S-7$, with audit as a marking on those edges rather than a new sphere.

The two drawings

Both are built in the RCN Graph Tool; the JSON files open there for editing. Hovering any node or line in the tool, or in the SVG files, shows the reasoning behind it.

	Crosswalk, strong match
	Crosswalk, partial match
	Ostrom variable carried by the informal OU-to-OU channels (lines into the dot marked T01)
	eVSM change agreed in this discussion
	eVSM channel, as drawn
☐ dashed	Added for the crosswalk: Next recursion up, Viability

Banded by level (next two pages): Ostrom on the left, the eVSM on the right, one band per level. It gives up the eVSM's geometric shape so the crosswalk lines never cross a node. The sphere-to-sphere mesh and environment couplings are left out; they are in the as-drawn view. The small italic line on each node gives its blanket role.

As drawn (the landscape page): both source diagrams with their own positions and links unchanged, joined by the crosswalk. It is busy on purpose: it shows where each source puts things. Each node's small italic lines give its IAD level and blanket role.

OSTROM / JONES 2003: conditions

eVSM FOR RCN: functions and environment

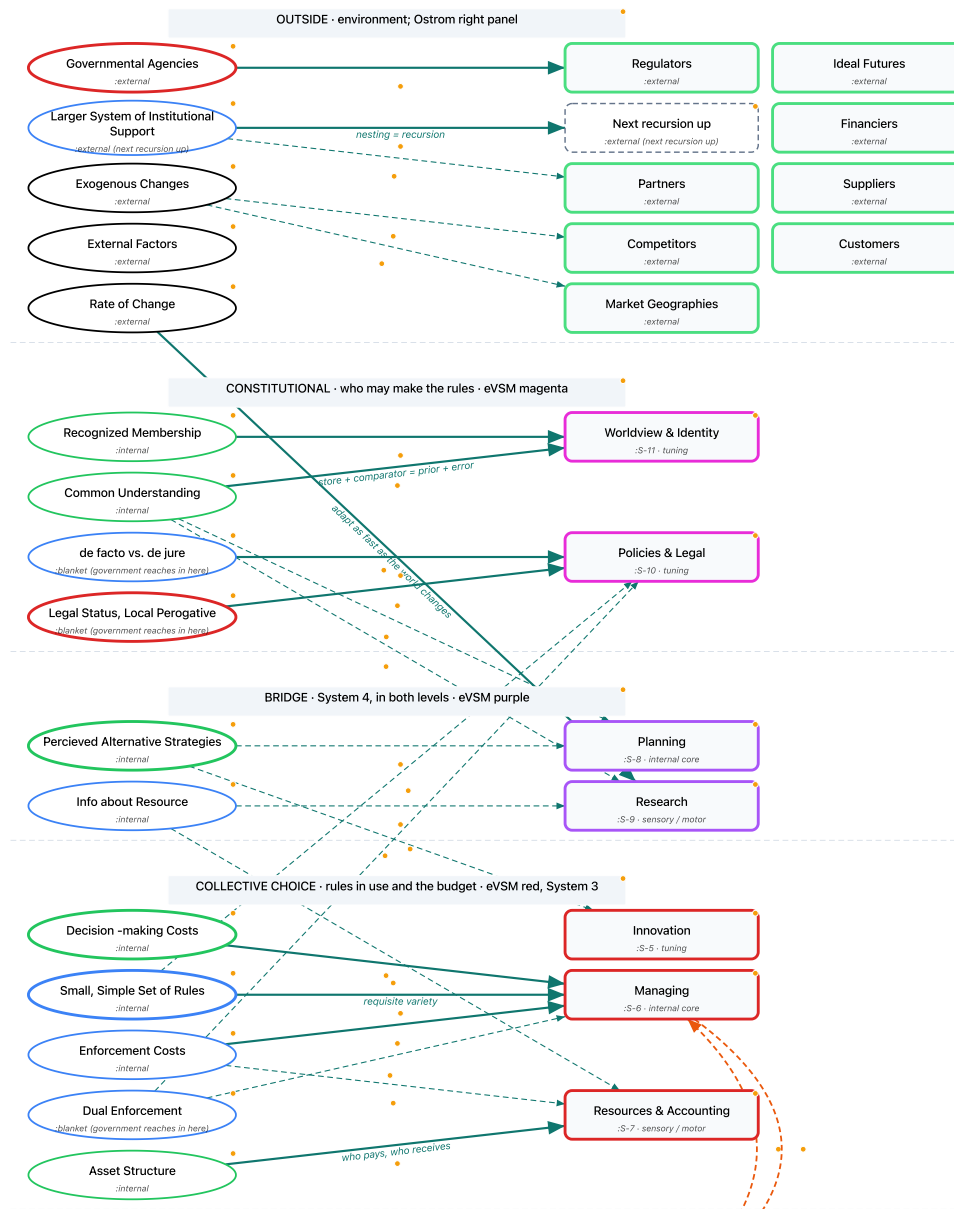


Figure 1a. Banded crosswalk: outside, constitutional, bridge and collective choice.

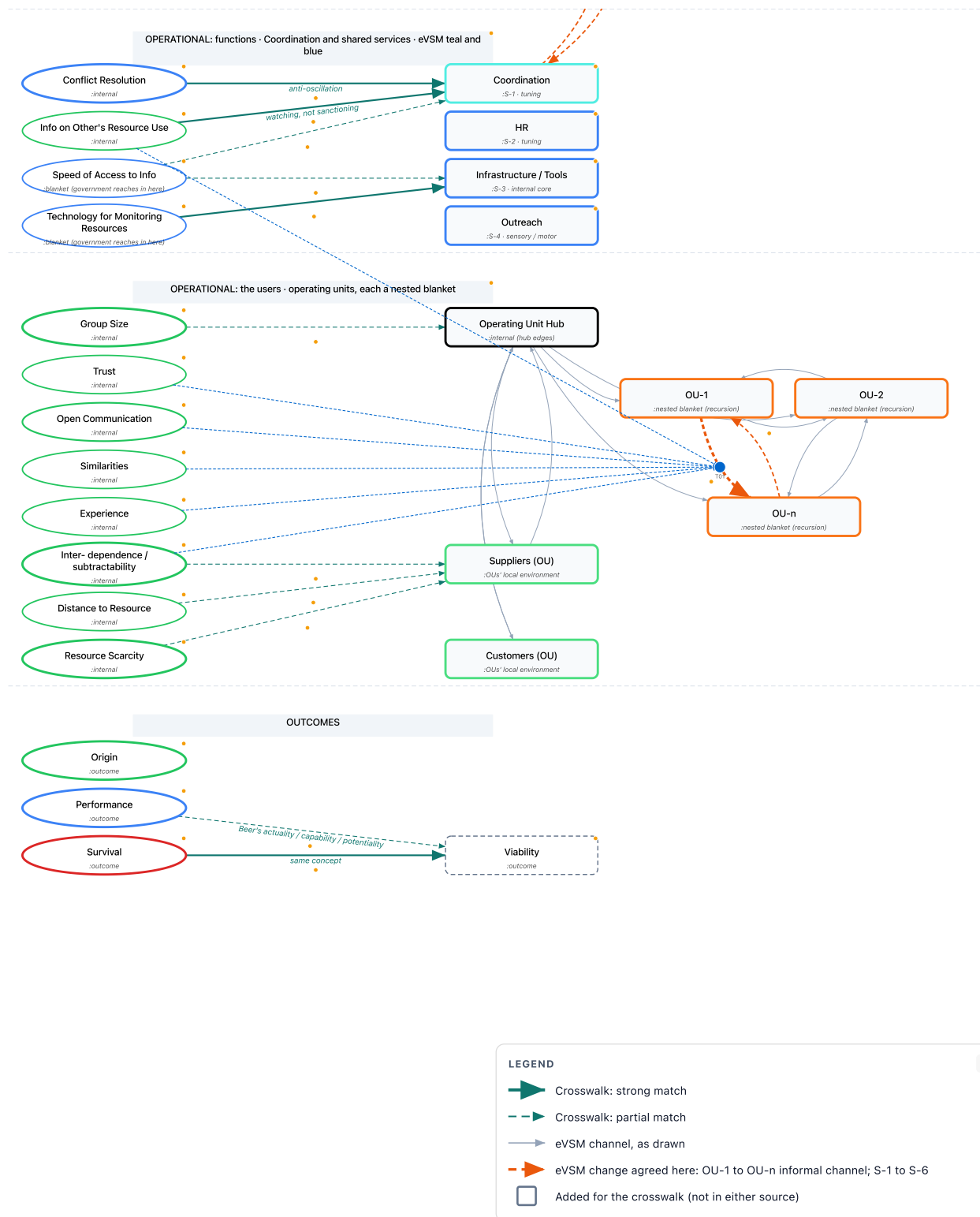


Figure 1b. Banded crosswalk: operational functions, the users and the outcomes. The orange S-1 to S-6 link continues from Figure 1a.

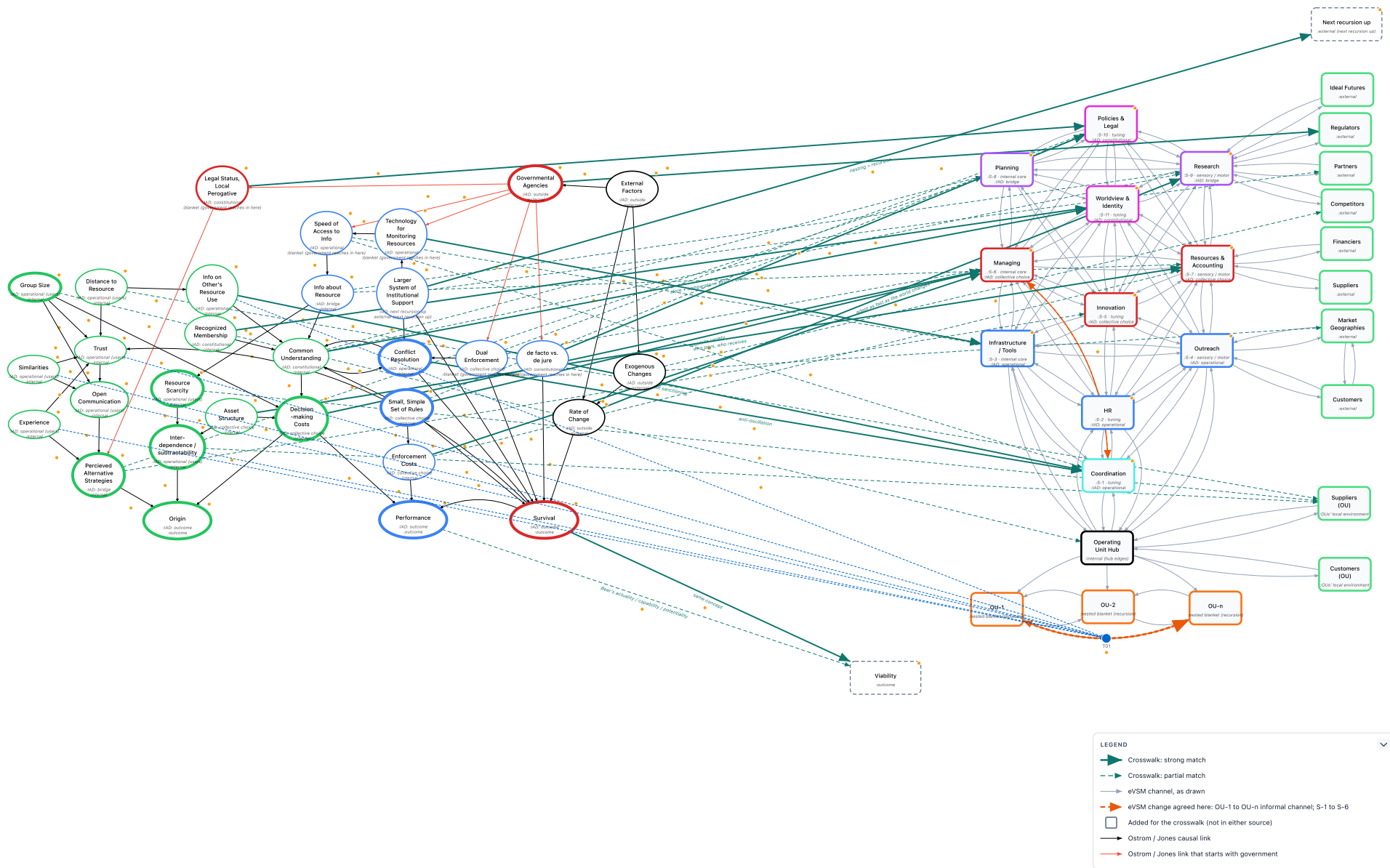


Figure 2. The two diagrams as drawn: Jones's redrawing of Ostrom (left), the eVSM plus environment (right), joined by the crosswalk.

Sources and files

- Jones, E. C. (2003). Building on Ostrom's "The Rudiments of a Theory of the Origins, Survival, and Performance of Common-property Institutions." *Journal of Ecological Anthropology* 7(1): 65–72, Figure 1. doi:10.5038/2162-4593.7.1.4
- Ostrom, E. (1990). *Governing the Commons*. Cambridge University Press (the design principles).
- Walker, J. (probably with A. Espinosa). "The Commons and the Work of Elinor Ostrom," section 7.2.5 of a longer work; PDF dated October 2019. Cites Espinosa & Walker (2013) and Espinosa et al. (2005).
- Beer, S. (1979, 1985). *The Heart of Enterprise; Diagnosing the System for Organizations*.
- RCN: docs/evsm-intro.html, docs/evsm-measures-spec.html.
- Graph Tool files, in docs/ostrom-evsm/: ostrom-evsm-banded.json and ostrom-evsm-as-drawn.json (editable), each with .svg and .png exports.